Name: ____________

Date: ____________

Function Behavior & Structure

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can determine whether a relation is a function.**

RELATED SKILLS
- D — Identify input and output values in a formula, graph, table, or context.
- A — Apply the vertical line test to a graph.

☐ **2. I can evaluate a function from a formula, graph, or table.**

RELATED SKILLS
- D — Identify input and output values in a formula, graph, table, or context.
- A — Simplify an algebraic expression using operation properties.
- A — Solve for an input associated with a specified output.

☐ **3. I can determine domain and range from a formula, graph, or table.**

RELATED SKILLS
- I — Determine domain restrictions from denominators, even roots, and logarithms.
- D — Read domain and range from a graph.
- D — Read domain and range from a table.

☐ **4. I can determine algebraic domain restrictions caused by denominators and even roots.**

RELATED SKILLS
- D — Determine domain restrictions from denominators, even roots, and logarithms.
- D — Require an even-root radicand to be nonnegative when determining domain.
- A — Determine excluded values from an original denominator.
- A — Factor a monic quadratic trinomial.
- A — Factor a quadratic trinomial with a nonunit leading coefficient.

☐ **5. I can graph a piecewise function from its formula.**

RELATED SKILLS
- I — Graph each rule on its stated domain with correct endpoints.
- I — Select the correct rule of a piecewise function for a given input.
- A — Describe increasing or decreasing behavior over intervals.

☐ **6. I can write a piecewise formula that represents a graph.**

RELATED SKILLS
- I — Translate a segmented graph into a piecewise formula.
- A — Convert among common forms of a linear equation.
- A — Describe increasing or decreasing behavior over intervals.

☐ **7. I can graph a transformed function using shifts, stretches, compressions, and reflections.**

RELATED SKILLS
- I — Distinguish stretches from compressions and horizontal effects from vertical effects.
- I — Interpret transformations applied to function inputs.
- I — Interpret transformations applied to function outputs.
- A — Connect parameters in a formula to features of its graph.

☐ **8. I can describe how a transformation affects domain, range, or asymptotes.**

RELATED SKILLS
- I — Exchange domain and range between a function and its inverse.
- D — Distinguish stretches from compressions and horizontal effects from vertical effects.
- D — Interpret transformations applied to function inputs.
- D — Interpret transformations applied to function outputs.

☐ **9. I can evaluate a composition from formulas, graphs, or tables.**

RELATED SKILLS
- I — Follow intermediate values to compose functions from tables.
- I — Read and write notation for function composition and inverse functions.
- I — Substitute one function formula into another.
- A — Identify input and output values in a formula, graph, table, or context.

☐ **10. I can write a formula for a composition of two functions.**

RELATED SKILLS
- D — Substitute one function formula into another.
- A — Simplify an algebraic expression using operation properties.

☐ **11. I can find an inverse function algebraically.**

RELATED SKILLS

- I — Exchange input and output values when interpreting an inverse.
- I — Exchange variables and solve to obtain an inverse formula.
- A — Isolate a specified variable in an equation or formula.
- A — Read and write notation for function composition and inverse functions.

☐ **12. I can evaluate and interpret an inverse from a graph, formula, or table.**

RELATED SKILLS

- D — Exchange input and output values when interpreting an inverse.
- A — Exchange variables and solve to obtain an inverse formula.
- A — Identify input and output values in a formula, graph, table, or context.

☐ **13. I can determine and interpret the units of an inverse function.**

RELATED SKILLS

- I — Reverse input and output units when interpreting an inverse.
- A — Attach and interpret units for rates, inputs, and outputs.

☐ **14. I can explain how the domain and range of a function relate to those of its inverse.**

RELATED SKILLS

- I — Exchange domain and range between a function and its inverse.
- A — Exchange input and output values when interpreting an inverse.

☐ **15. I can determine the range of a function by analyzing the domain of its inverse.**

RELATED SKILLS

- D — Exchange domain and range between a function and its inverse.
- A — Determine domain restrictions from denominators, even roots, and logarithms.

☐ **16. I can verify algebraically that two functions are inverses.**

RELATED SKILLS

- I — Verify inverse functions using composition.
- A — Simplify an algebraic expression using operation properties.
- A — Substitute one function formula into another.

☐ **17. I can interpret the meaning and units of a composition.**

RELATED SKILLS

- D — Attach and interpret units for rates, inputs, and outputs.
- D — Follow intermediate values to compose functions from tables.
- D — Substitute one function formula into another.
- A — Explain a model parameter in the language of its context.

☐ **18. I can classify a graph or table as concave up or concave down using changes in average rate of change.**

RELATED SKILLS

- I — Use successive average rates of change to classify concavity.
- D — Use tabular patterns to predict graph shape and behavior.
- A — Calculate average rate of change over an interval.

☐ **19. I can determine whether a function is invertible using the horizontal line test.**

RELATED SKILLS

- I — Apply the horizontal line test for invertibility.
- I — Test whether a function is one-to-one on a domain.

☐ **20. I can graph an inverse function as a reflection across the line y = x.**

RELATED SKILLS

- D — Connect parameters in a formula to features of its graph.
- D — Exchange input and output values when interpreting an inverse.
- A — Exchange domain and range between a function and its inverse.